A direct approach to computing the non-interacting kinetic energy functional

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21st April, 2026

Outline

- Motivation
- Research Gap
- Proposed Theory
- Numerical Validation
- Numerical study
- Analytical Study
- Summary

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Energy of a system of atoms

Under the Born-Oppenheimer approximation:

$$I(R_1, R_2, ..., R_M) = \inf_{\substack{\psi \in \mathcal{Q} \\ \int_X |\psi|^2 = 1}} (\psi, H_v \psi) + \sum_{\substack{i,j=1 \\ i<j}}^{M} \frac{Z_i Z_j}{\|R_i - R_j\|},$$

$$H_v = -\frac{1}{2}\sum_{i=1}^{N}(\Delta_i + v(r_i)) + \sum_{\substack{i,j=1 \\ i<j}}^{N} \frac{1}{\|r_i - r_j\|}; \quad v(r_i) = -\sum_{j=1}^{M} \frac{Z_j}{\|r_i - R_j\|},$$

$$\mathcal{Q} = \{\psi \in \wedge_1^N L^2(X) : \sum_{i=1}^{N} \int_X |\nabla_{r_i} \psi|^2 < \infty\}; \quad X = (\mathbb{R}^d \times \mathbb{Z}_2)^N.$$

Ground state

- $Y^d = L^p(\mathbb{R}^d;\mathbb{R}) + L^\infty(\mathbb{R}^d;\mathbb{R})$ with

$$p \begin{cases} = 1 & \text{when } d = 1, \\ > 1 & \text{when } d = 2, \\ = \frac{d}{2} & \text{when } d \geq 3. \end{cases}$$

- Ground state energy, E, for a given potential $v \in Y^d$ is defined as

$$E[v] := \inf_{\substack{\psi \in \mathcal{Q} \\ \int_X |\psi|^2 = 1}} (\psi, H_v \psi)\,.$$

Electron density

- For a given $\psi \in \mathcal{Q}$ define electron density, ρ_ψ, and then define a set of $N-$representable densities, $\mathcal{I}_N$, as

$$\rho_\psi(r) := N \sum_\sigma \int |\psi(r, \sigma_1, r_2, \sigma_2, ..., r_N, \sigma_N)|^2 dr_2 dr_3 ... dr_N.$$

$$\mathcal{I}_N := \{\rho \in L^1(\mathbb{R}^d, \mathbb{R}^+) : \sqrt{\rho} \in H^1(\mathbb{R}^d), \int_{\mathbb{R}^d} \rho = N\}.$$

- It is easy to see that $\rho_\psi \in \mathcal{I}_N$ when $\int_X |\psi|^2 = 1$. Let $\rho \in \mathcal{I}_N$, then there exists a $\psi \in \mathcal{Q}$ with $\int_X |\psi|^2 = 1$ such that $\rho_\psi = \rho$.

$$\boxed{E[v] = \inf_{\substack{\psi \in \mathcal{Q} \\ \int_X |\psi|^2 = 1}} (\psi, H_v \psi) = \inf_{\rho \in \mathcal{I}_N} \inf_{\substack{\psi \in \mathcal{Q} \\ \int_X |\psi|^2 = 1}} \{(\psi, H_v \psi) : \rho_\psi = \rho\}}$$

Density Functional Theory (DFT)

$$E[v] = \inf_{\rho \in \mathcal{I}_N} \inf_{\substack{\psi \in \mathcal{Q} \\ \int_X |\psi|^2 = 1}} \{(\psi, H_v \psi) : \rho_\psi = \rho\},$$

$$H_v = -\frac{1}{2}\left(\sum_{i=1}^{N} \Delta_i + v(r_i)\right) + \sum_{\substack{i,j=1 \\ i<j}}^{N} \frac{1}{\|r_i - r_j\|}.$$

- Due to Hohenberg and Kohn,

$$E[v] = \inf_{\rho \in \mathcal{I}_N} \left(F(\rho) + \int \rho v\right),$$

$$F(\rho) = \inf_{\substack{\psi \in \mathcal{Q} \\ \int_X |\psi|^2 = 1}} \{(\psi, H_0 \psi) : \rho_\psi = \rho\} = \min_{\substack{\psi \in \mathcal{Q} \\ \int_X |\psi|^2 = 1}} \{(\psi, H_0 \psi) : \rho_\psi = \rho\}.$$

Kohn-Sham DFT

$$E[v] = \inf_{\rho \in \mathcal{I}_N} \left(F(\rho) + \int \rho v \right),$$

$$F(\rho) = \min_{\substack{\psi \in \mathcal{Q} \\ \int_X |\psi|^2 = 1}} \{ (\psi, H_0 \psi) : \rho_\psi = \rho \}.$$

$$\psi(x_1, \ldots, x_n) = \frac{1}{\sqrt{n!}} \begin{vmatrix} \phi_1(x_1) & \phi_2(x_1) & \cdots & \phi_n(x_1) \\ \phi_1(x_2) & \phi_2(x_2) & \cdots & \phi_n(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(x_n) & \phi_2(x_n) & \cdots & \phi_n(x_n) \end{vmatrix}.$$

Let $\rho \in \mathcal{I}_N$. Then,

$$F(\rho) = T_{KS}(\rho) + E_{xc}(\rho) + \frac{1}{2}\int\int \frac{\rho(x)\rho(y)}{|x-y|},$$

$$T_{KS}(\rho) = \frac{1}{2} \min_{\substack{\phi_1,\phi_2,..,\phi_N \in H^1(\mathbb{R}^d \times \mathbb{Z}_2, \mathbb{C}) \\ \left(\phi_i,\phi_j\right)_{L^2(\mathbb{R}^d,\mathbb{C})} = \delta_{ij} \\ \sum_{i=1}^N \sum_{\sigma\in\mathbb{Z}_2} |\phi_i(\cdot,\sigma)|^2 = \rho(\cdot)}} \sum_{i=1}^N \int_X |\nabla \phi_i|^2$$

$$E[v] = \inf_{\rho \in \mathcal{I}_N} \left(F(\rho) + \int \rho v \right)$$

Differential form

$$\frac{\delta T_{KS}}{\delta \rho} + \left(v + U_H + \frac{\delta E_{xc}}{\delta \rho} \right) = \mu.$$

$$v_{KS} = v + U_H + \frac{\delta E_{xc}}{\delta \rho}.$$

$$-\frac{1}{2}\nabla^2 \phi_i + v_{KS}\, \phi_i = \epsilon_i \phi_i,$$

$$\rho(r) = \sum_i \sum_{s=\{\uparrow,\downarrow\}} |\phi_i(r,s)|^2.$$

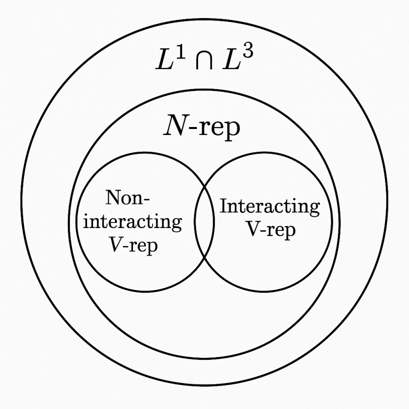
$$T_{KS}(\rho) = \frac{1}{2} \sum_{i=1}^{N} \int_X |\nabla \phi_i|^2$$

Non-interacting v-representable

$$\mathcal{P}_N = \{\Phi = (\phi_1, \phi_2, ..., \phi_N) \in H^1(\mathbb{R}^d; \mathbb{R}^N) \ : \ (\phi_i, \phi_j)_{L^2(\mathbb{R}^d)} = \delta_{ij}, \\ \forall 1 \leq i, j \leq N\}.$$

$$V_N = \left\{ v \in Y^d : \exists\, \Phi^\circ \in \mathcal{P}_N \text{ s.t. } \Phi^\circ = \arg\inf_{\Phi \in \mathcal{P}_N} \sum_{i=1}^{N} \int_{\mathbb{R}^d} \left(\|\nabla \phi_i\|^2 + 2\phi_i^2 v \right) \right\}.$$

Space of electron density



Kinetic energy functionals

$$T_{\mathrm{LL}}[\rho] := \inf_{\substack{\Psi \in \bigwedge_1^N L^2(\mathbb{R}^d \times \mathbb{Z}_2, \mathbb{C}) \\ \|\Psi\|_{L^2}^N = 1 \\ \rho_\Psi = \rho}} \frac{1}{2} \sum_{j=1}^{N} \int_{(\mathbb{R}^d \times \mathbb{Z}_2)^N} |\nabla_j \Psi(\mathbf{X})|^2 \, \mathrm{d}\mathbf{X}.$$

$$T_{\mathrm{L}}[\rho] := \inf_{\substack{\Gamma = \Gamma^* \geqslant 0 \\ \operatorname{Tr} \Gamma = 1 \\ \operatorname{Tr}(-\Delta)\Gamma < \infty \\ \rho_\Gamma = \rho}} \operatorname{Tr}\left(H_N^{0,0} \Gamma\right) = \inf_{\substack{\gamma = \gamma^\dagger \\ 0 \leq \gamma \leq 1 \\ \operatorname{Tr} \gamma = N \\ \gamma(r,r) = \rho(r)}} \operatorname{Tr}(-\Delta\, \gamma)\,.$$

$$= \sup_{v \in L^p(\mathbb{R}^d) + L^\infty(\mathbb{R}^d)} \left\{ E_N[v] - \int_{\mathbb{R}^d} v(\mathbf{r}) \rho(\mathbf{r}) \mathrm{d}\mathbf{r} \right\}$$

$$T_{det}[\rho] := \min_{\substack{\varphi_1, \ldots, \varphi_N \in H^1(\mathbb{R}^d \times \mathbb{Z}_q, \mathbb{C}) \\ \langle \varphi_i, \varphi_j \rangle = \delta_{ij} \\ \rho_\Phi = \rho}} \frac{1}{2} \sum_{j=1}^{N} \int_{\mathbb{R}^d \times \mathbb{Z}_q} |\nabla \varphi_j(\mathbf{x})|^2 \, \mathrm{d}\mathbf{x}$$

Why OFDFT?

- Optimization over N functions
- Typical algorithm scales cubically with N

$$F(\rho) = T_{KS}(\rho) + E_{xc}(\rho) + \frac{1}{2}\int\int \frac{\rho(x)\rho(y)}{|x-y|}.$$

Goal

- *How to accurately and cost-effectively calculate the non-interacting kinetic energy functional?*

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Restricted Kohn-Sham setting

Let $\rho \in \mathcal{I}_{2N}$. Then,

$$T_{RKS}(\rho) = \min_{\substack{\phi_1,\phi_2,..,\phi_n \in H^1(\mathbb{R}^d) \\ \left(\phi_i,\phi_j\right)_{L^2(\mathbb{R}^d)}=\delta_{ij} \\ \sum_{i=1}^n |\phi_i|^2=\rho}} \sum_{i=1}^{N} \int_{\mathbb{R}^d} |\nabla \phi_i|^2$$

Data based approach

$v_{KS} \to \{\phi_1, \phi_2, ..., \phi_N\}$ or Inverse Kohn-Sham

$$\rho = 2\sum_{i=1}^{N} \phi_i^2,$$

$$T_{RKS}(\rho) = \sum_{i=1}^{n} \int_{\mathbb{R}^3} |\nabla \phi_i|^2.$$

$$\boxed{E[v] = \inf_{\rho \in \mathcal{I}_N} \left(F(\rho) + \int \rho v \right)}$$

$$\mathcal{I}_N = \{\rho \in L^1(\mathbb{R}^d, \mathbb{R}^+) : \sqrt{\rho} \in H^1(\mathbb{R}^d), \int_{\mathbb{R}^d} \rho = N\}$$

Objectives

- Extend the definition of the determinant non-interacting kinetic energy functional
- Explicitly and accurately calculate the non-interacting kinetic energy functional

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$$\mathcal{I}_N = \{\rho \in L^1(\mathbb{R}^d) : \rho \geq 0, \sqrt{\rho} \in H^1(\mathbb{R}^d), \int_{\mathbb{R}^d} \rho = N\}$$

$$T_{RKS}(\rho) = \min_{\mathcal{Q}_\rho^{++}} \sum_{i=1}^{N} \int_{\mathbb{R}^d} |\nabla \phi_i|^2,$$

$$\mathcal{Q}_\rho^{++} = \{(\phi_1, , ..., \phi_N) \in H^1(\mathbb{R}^d;\mathbb{R}^N), (\phi_i, \phi_j)_{L^2(\mathbb{R}^d)} = \delta_{ij}, 2\sum_{i=1}^{N} |\phi_i|^2 = \rho\}.$$

Extension of T_{KS}

- Define an natural number N_u as ceiling of $\int_{\mathbb{R}^d} \frac{u^2}{2}$ for any $u \in L^2(\mathbb{R}^d)$.
- Define $L : H^1(\mathbb{R}^d; \mathbb{R}^{N_u}) \to \mathbb{R}^+$ be such that

$$L(\phi_1, \phi_2, ..., \phi_{N_u}) = \sum_{i=1}^{N_u} \int_{\mathbb{R}^d} |\nabla \phi_i|^2.$$

$$\mathcal{M}_u = \{(\phi_1, \phi_2, ..., \phi_{N_u}) \in H^1(\mathbb{R}^d; \mathbb{R}^{N_u}) | (\phi_i, \phi_j)_{L^2(\mathbb{R}^d)} = 0$$
$$\forall i, j \in [1, N_u] \quad \text{st} \quad i \neq j,$$
$$\|\phi_i\|_{L^2(\mathbb{R}^d)} = 1 \quad \forall i \in [1, N_u - 1], \quad 2\sum_{i=1}^{N_u} |\phi_i|^2 = u^2\}.$$

Lemma

Let $u \in H^1(\mathbb{R}^d)$ such that $N_u \geq 1$. Then there exists $\Phi \in \mathcal{M}_u$ such that $L(\Phi) \leq C \int_{\mathbb{R}^d} \|\nabla u\|^2$ for some positive constant, C (dependent on N_u). Moreover, for any $\Phi \in \mathcal{M}_u$, $\int_{\mathbb{R}^d} \|\nabla u\|^2 \leq 2L(\Phi)$.

Define $T : H^1(\mathbb{R}^d) \to \mathbb{R}_{\geq 0}$ such that

$$T(u) = \begin{cases} 0, & N_u = 0, \\ \inf_{\Phi \in \mathcal{M}_u} L(\Phi), & N_u \geq 1. \end{cases} \tag{1}$$

Theorem

Let $u \in H^1(\mathbb{R}^d)$ such that $N_u \geq 1$. Then infima in 1 is achieved.

$$T(\sqrt{\rho}) = T_{RKS}(\rho) \quad \forall \rho \in \mathcal{I}_N.$$

Issue in Minimization

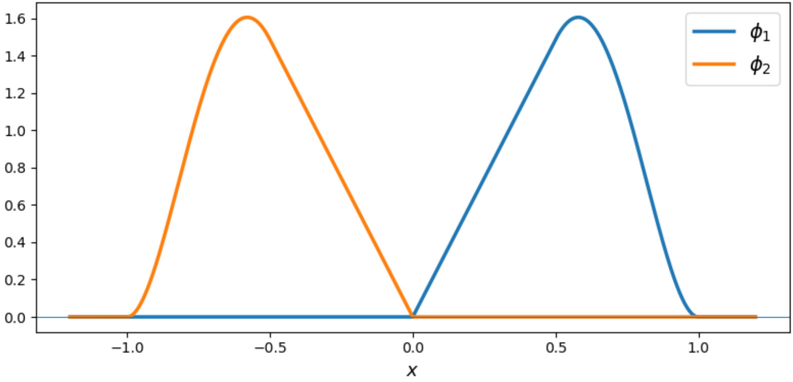
$$T(u) = \min_{\Phi \in \mathcal{M}_u} L(\Phi).$$

$$\begin{aligned}\mathcal{M}_u = \{(\phi_1, \phi_2, ..., \phi_{N_u}) \in H^1(\mathbb{R}^d; \mathbb{R}^{N_u}) | (\phi_i, \phi_j)_{L^2(\mathbb{R}^d)} = 0 \\ \forall i, j \in [1, N_u] \quad \text{st} \quad i \neq j, \\ \|\phi_i\|_{L^2(\mathbb{R}^d)} = 1 \quad \forall i \in [1, N_u - 1], \quad 2\sum_{i=1}^{N_u} |\phi_i|^2 = u^2\}.\end{aligned}$$

Adjoint for $u^2 = 2\sum_{i=1}^{N_u} \phi_i^2$? Singularity in adjoint?

$$
\begin{aligned}
\phi_1 &= \frac{|u|}{\sqrt{2}} \sin\theta_1, \\
\phi_2 &= \frac{|u|}{\sqrt{2}} \cos\theta_1 \sin\theta_2, \\
\phi_3 &= \frac{|u|}{\sqrt{2}} \cos\theta_1 \cos\theta_2 \sin\theta_3, \\
&\vdots \\
\phi_k &= \frac{|u|}{\sqrt{2}} \cos\theta_1 \cos\theta_2 \cdots \cos\theta_{k-1} \sin\theta_k, \\
&\vdots \\
\phi_{N-1} &= \frac{|u|}{\sqrt{2}} \cos\theta_1 \cos\theta_2 \cdots\cdots \cos\theta_{N-2} \sin\theta_{N-1}, \\
\phi_N &= \frac{|u|}{\sqrt{2}} \cos\theta_1 \cos\theta_2 \cdots\cdots \cos\theta_{N-2} \cos\theta_{N-1}.
\end{aligned}
$$

Discontinuity due to inversion



Thus $\{\phi_1, \phi_2\}$ forms an orthonormal set in $L^2(\mathbb{R})$. Setting

$$u(x) = \sqrt{2\phi_1(x)^2 + 2\phi_2(x)^2} \tag{2}$$

we may formally write

$$\phi_1(x) = \frac{u(x)}{\sqrt{2}} \sin(\theta(x)), \quad \phi_2(x) = \frac{u(x)}{\sqrt{2}} \cos(\theta(x)). \tag{3}$$

Note that $u \in H^1(\mathbb{R})$. Pointwise, the angle θ is determined as

$$\theta(x) = \begin{cases} 0, & x < 0, \\ \pi/2, & x > 0, \end{cases} \tag{4}$$

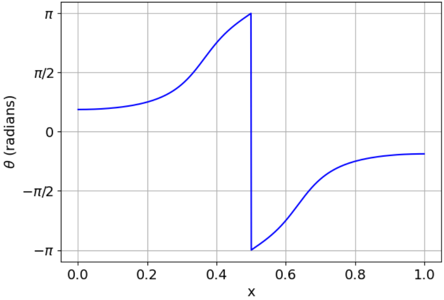
with any arbitrary value at $\theta(0)$. Hence θ has jump discontinuity at $x = 0$ and is not in $H^1_{loc}(\mathbb{R})$.

$$
\begin{aligned}
\phi_1 &= \frac{|u|}{\sqrt{2}} \sin\theta_1, \\
\phi_2 &= \frac{|u|}{\sqrt{2}} \cos\theta_1 \sin\theta_2, \\
\phi_3 &= \frac{|u|}{\sqrt{2}} \cos\theta_1 \cos\theta_2 \sin\theta_3, \\
&\vdots \\
\phi_k &= \frac{|u|}{\sqrt{2}} \cos\theta_1 \cos\theta_2 \cdots \cos\theta_{k-1} \sin\theta_k, \\
&\vdots \\
\phi_{N-1} &= \frac{|u|}{\sqrt{2}} \cos\theta_1 \cos\theta_2 \cdots\cdots \cos\theta_{N-2} \sin\theta_{N-1}, \\
\phi_N &= \frac{|u|}{\sqrt{2}} \cos\theta_1 \cos\theta_2 \cdots\cdots \cos\theta_{N-2} \cos\theta_{N-1}.
\end{aligned}
$$

$$
\operatorname{atan2}(y, x) = \begin{cases} \arctan\left(\dfrac{y}{x}\right), & x > 0, \\ \arctan\left(\dfrac{y}{x}\right) + \pi, & y \geq 0,\ x < 0, \\ \arctan\left(\dfrac{y}{x}\right) - \pi, & y < 0,\ x < 0, \\ +\dfrac{\pi}{2}, & y > 0,\ x = 0, \\ -\dfrac{\pi}{2}, & y < 0,\ x = 0, \\ \text{undefined}, & y = 0,\ x = 0. \end{cases}
$$

Discontinuity due to incompatibility

- Let $\phi_1(x) = \sqrt{2}\sin(2\pi x), \quad \phi_2(x) = \sqrt{2}\sin(3\pi x)$ and define that $2\phi_1^2 + 2\phi_2^2 = u^2$. Note that $(\phi_1, \phi_2) \in \mathcal{M}_u$.
- $\phi_1 = \frac{|u|}{2}\sin(\theta), \quad \phi_2 = \frac{|u|}{2}\cos(\theta)$.
- θ is uniquely determined as $\theta = \text{atan2}(\phi_2, \phi_1)$.

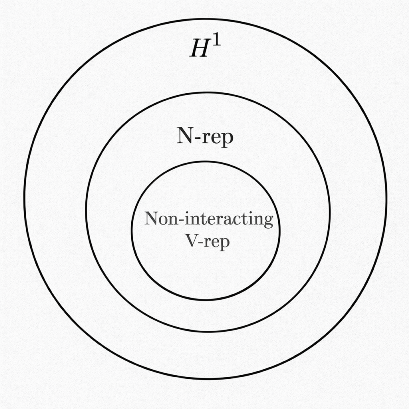


$$\operatorname{atan2}(y,x) = \begin{cases} \arctan\left(\dfrac{y}{x}\right), & x > 0, \\ \arctan\left(\dfrac{y}{x}\right) + \pi, & y \geq 0,\ x < 0, \\ \arctan\left(\dfrac{y}{x}\right) - \pi, & y < 0,\ x < 0, \\ +\dfrac{\pi}{2}, & y > 0,\ x = 0, \\ -\dfrac{\pi}{2}, & y < 0,\ x = 0, \\ \text{undefined}, & y = 0,\ x = 0. \end{cases}$$

$$T^+(u) = \min_{\Phi \in \mathcal{M}_u^+} L(\Phi)$$
$$\mathcal{M}_u^+ = \{\Phi = (\phi_1, \phi_2, ..., \phi_{N_u}) \in \mathcal{M}_u \ : \ \phi_{N_u} \geq 0\}.$$

- Since $\mathcal{M}_u^+ \subset \mathcal{M}_u$, it holds that $T(u) \leq T^+(u)$.
- $T(u) = T^+(u)$ when u^2 is a non-interacting v-representable ground state density.
- Both T and T^+ are continuous.

Square root of electron density



Parameterization

- Let $u \in H^1(\mathbb{R}^d)$ such that $N_u \geq 2$.
- For $\Theta = (\theta_1, \theta_2, ..., \theta_{N-1}) \in \mathcal{S}_u$ construct $\varphi_N^{\Theta,u} = \frac{|u|}{\sqrt{2}} \cos\theta_1 \cos\theta_2 ..... \cos\theta_{N-1}$ and

$$\varphi_k^{\Theta,u} = \frac{|u|}{\sqrt{2}} \cos\theta_1 \cos\theta_2 .... \cos\theta_{k-1} \sin\theta_k = u_k \sin\theta_k$$

for all $1 \leq k \leq N-1$.

$$\boxed{\begin{aligned} \mathcal{S}_u = \{(\theta_1, \theta_2, ..., \theta_{N_u-1}) \in H^1_{loc}(\mathbb{R}^d; \mathbb{R}^{N-1}) \quad | \\ u_k\theta_i \in H^1(\mathbb{R}^d)) \quad \forall i \in [1, N-1]\}. \end{aligned}}$$

- This implies $|\varphi_1^{\Theta,u}|^2 + ... + |\varphi_N^{\Theta,u}|^2 = \frac{u^2}{2}$.
- Then we show $\varphi_k^{\Theta,u} \in H^1(\mathbb{R}^d;\mathbb{R})$ $1 \le k \le N$.
- $\mathcal{S}_u \cap A_u$ is non-empty and hence for every $\Theta \in \mathcal{S}_u \cap A_u$ there exists a $\Phi^{\Theta,u} = (\varphi_1^{\Theta,u}, \varphi_2^{\Theta,u}, ..., \varphi_N^{\Theta,u}) \in \mathcal{M}_u$ where

$$A_u = \{\Theta \in \mathcal{S}_u || \langle \varphi_i^{\Theta,u}, \varphi_j^{\Theta,u} \rangle_{L^2(R^d,\mathbb{R})} = 0 \quad \forall i,j \in [1,N] \quad \text{st} \quad i \neq j, \\ \|\varphi_i^{\Theta}\|_{L^2(\mathbb{R}^d)} = 1 \quad \forall i \in [1, N-1]\}.$$

Define $\tilde{L} : \mathcal{S}_u \to \mathbb{R}^+$ as follows:

$$\tilde{L}(\Theta) = L(\Phi^{\Theta,u}) = \sum_{k=1}^{N_u} \int_{\mathbb{R}^d} \|\nabla \varphi_k^{\Theta,u}\|^2.$$

Using induction it follows that

$$\tilde{L}(\Theta) = \frac{1}{2}\int_{\mathbb{R}^d} \|\nabla u\|^2 + \frac{1}{2}\int_{\mathbb{R}^d} u^2 \|\nabla \theta_1\|^2 + \frac{1}{2}\int_{\mathbb{R}^d} u^2 \Big(\cos^2\theta_1 \|\nabla \theta_2\|^2$$
$$+ \cos^2\theta_1 \cos^2\theta_2 \|\nabla \theta_3\|^2 + \ldots + \cos^2\theta_1 \ldots . \cos^2\theta_{N_u-2} \|\nabla \theta_{N_u-1}\|^2 \Big).$$

Minimisation Problem

Theorem

Let $u \in H^1(\mathbb{R}^d)$ with $N_u \geq 2$ and define a subset of $\mathcal{S}_u$ as follows:

$$\mathcal{S}_u^+ = \{\Theta \in \mathcal{S}_u \ : \ |\theta_i| \leq \pi/2, \ 1 \leq i < N_u\}.$$

Then,

$$T^+(u) = \lim_{\delta \downarrow 0^+} \inf_{\Theta \in \mathcal{S}_u^+ \cap \mathcal{A}_u^\delta} \tilde{L}(\Theta), \tag{5}$$

where, for every $\delta > 0$,

$$\mathcal{A}_u^\delta = \{\Theta \in L^1_{loc}(\mathbb{R}^d;\mathbb{R}^{N_u-1}) \ : \ |\langle \varphi_i^{\Theta,u}, \varphi_j^{\Theta,u} \rangle_{L^2(R^d,\mathbb{R})}| \leq \delta, \, 1 \leq i < j \leq N_u, \, |\|\varphi_i^{\Theta,u}\|_{L^2(\mathbb{R}^d)} - 1| \leq \delta, \, 1 \leq i < N_u\}.$$

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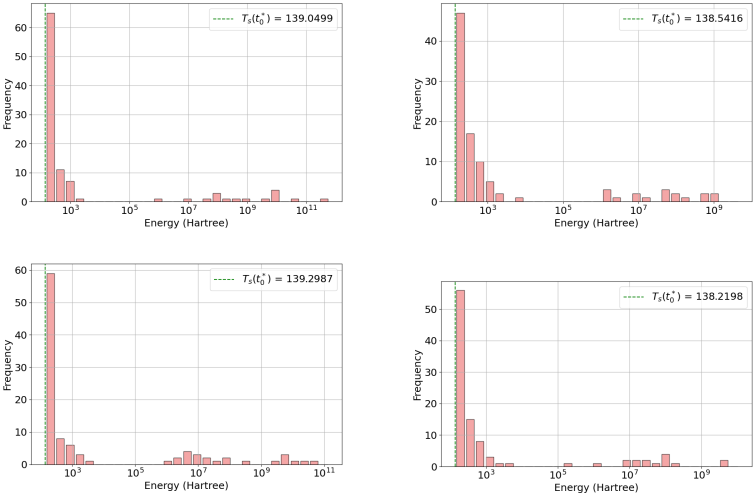
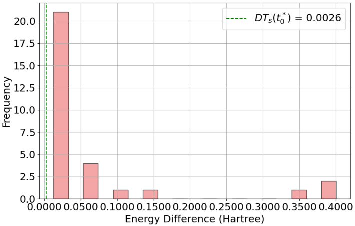
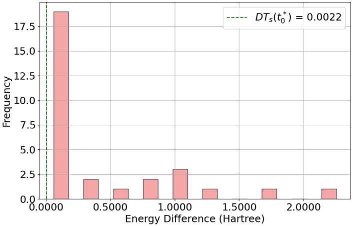
Problem Setting

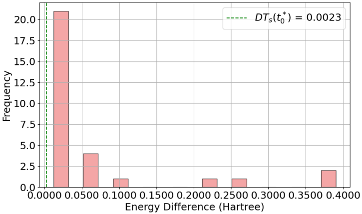
- We restrict ourselves to the 1D case on the domain $(0, 1)$.
- The SLSQP algorithm is used to solve the minimization problem and radial basis functions are used to approximate θ's.
- We set δ to be machine epsilon of ease of calculation.
- Since a local minimization method is used, we generate results from multiple initial conditions and select the lowest minimum as the final solution.

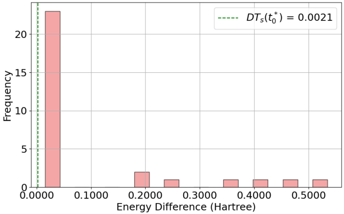
Special choice of Initial Condition




$DT_s(t_0^*) = 0.0026$
Frequency
Energy Difference (Hartree)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0.0000 0.0500 0.1000 0.1500 0.2000 0.2500 0.3000 0.3500 0.4000


$DT_s(t_0^*) = 0.0022$
Frequency
Energy Difference (Hartree)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0.0000 0.5000 1.0000 1.5000 2.0000


$DT_s(t_0^*) = 0.0023$
Frequency
Energy Difference (Hartree)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0.0000 0.0500 0.1000 0.1500 0.2000 0.2500 0.3000 0.3500 0.4000


$DT_s(t_0^*) = 0.0021$
Frequency
Energy Difference (Hartree)
0
5
10
15
20
0.0000 0.1000 0.2000 0.3000 0.4000 0.5000

1D Kohn Sham system

- Generated random 1D potential v.
- Compute the first N eiegnvectors of the operator

$$-\frac{1}{2}\Delta + v$$

  on the domain $(0,1)$ with Dirichlet boundary conditions.
- Compute electron density and kinetic energy

$$\rho = 2\sum_{i=1}^{N} \phi_i^2,$$

$$T_{KS}(\rho) = \sum_{i=1}^{N} \int_{\mathbb{R}^3} |\nabla \phi_i|^2.$$

$N = 2$

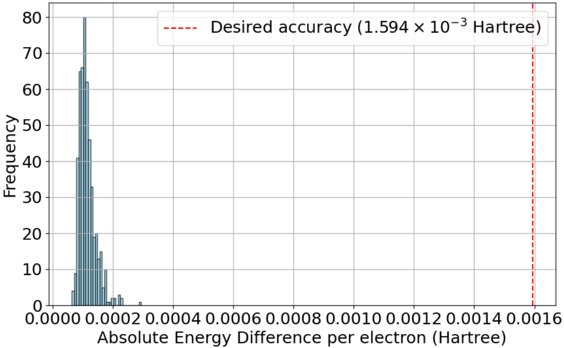


Figure 3: Error between the actual solution and the obtained minimum. Here $n_e = 1000$ and $n_{rbf} = 15$ is used.

$N = 3$

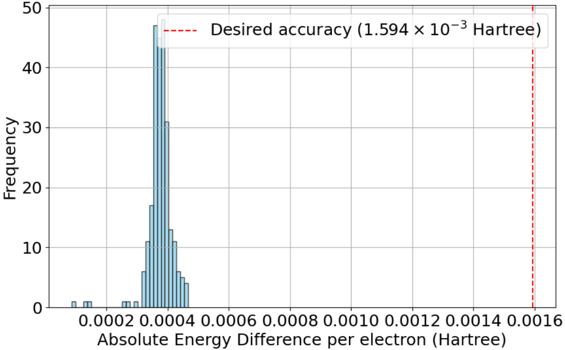


Figure 4: Error between the actual solution and the obtained minimum. Here $n_e = 1000$ and $n_{rbf} = 20$ is used.

$N = 4$

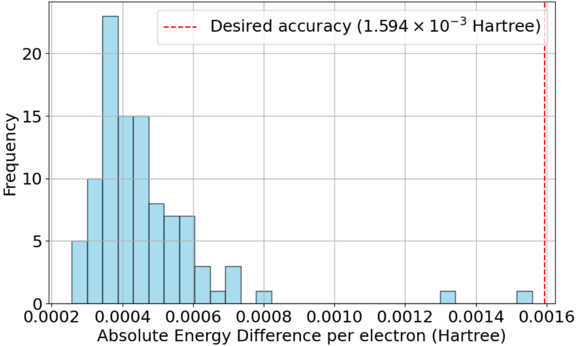


Figure 5: Error between the actual solution and the obtained minimum. Here $n_e = 2000$ and $n_{rbf} = 70$ is used.

Outline

- Motivation
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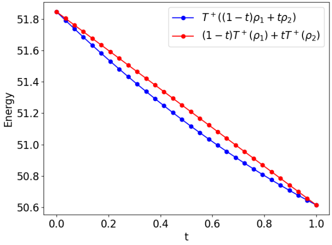
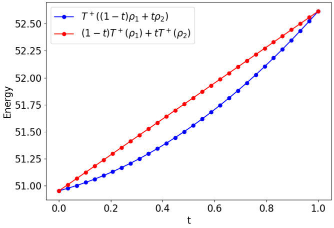
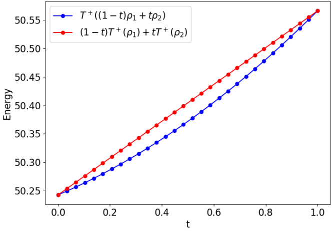
Investigation of properties of T^+

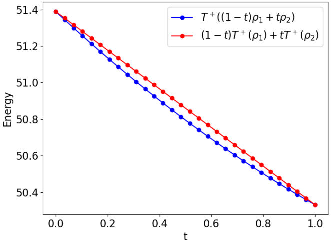
- We use ρ instead of u as the primary variable of T^+, but retain the functional symbol for notational convenience.
- Let ρ_1 and ρ_2 be two one-dimensional Kohn-Sham densities.
- We define a linear interpolation between them as $\rho_t = (1-t)\rho_1 + t\rho_2$ for all $t \in [0,1]$.

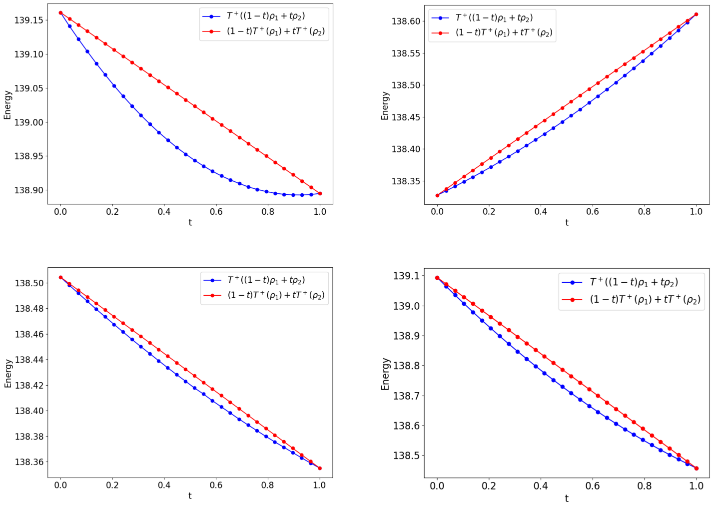
$\int \rho_1 = \int \rho_2 = 2N$ for $N = 2, 3, 4$

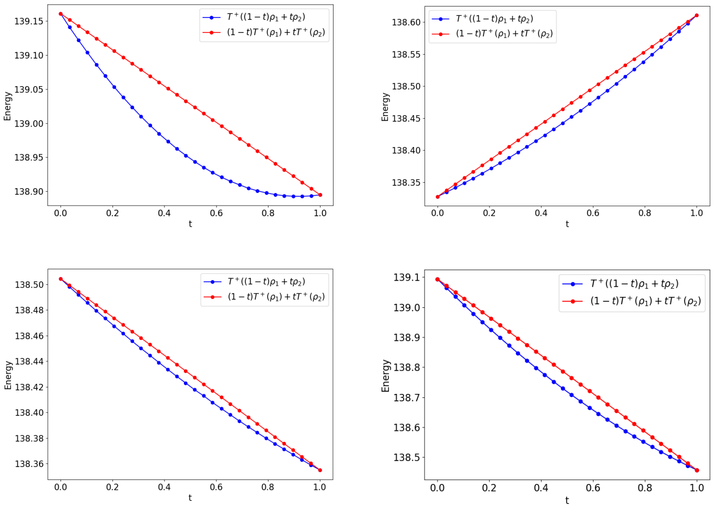




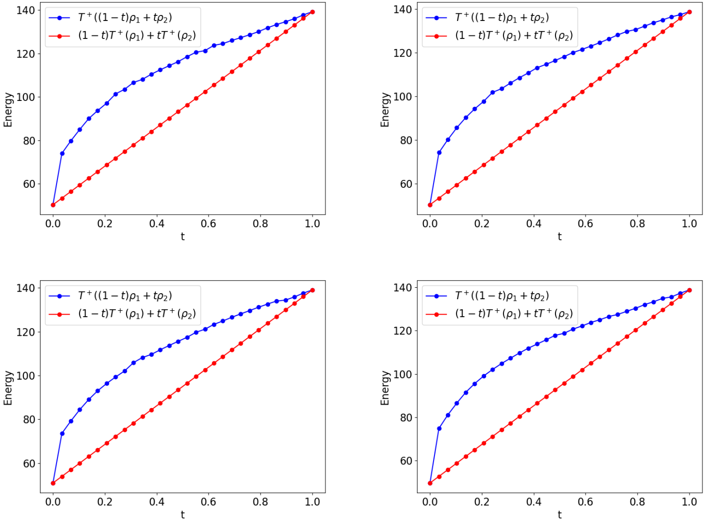




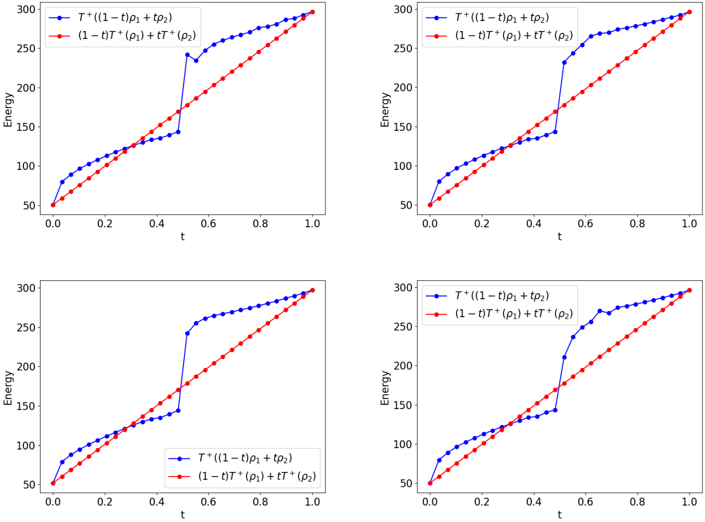

$T^+((1-t)\rho_1 + t\rho_2)$
$(1-t)T^+(\rho_1) + tT^+(\rho_2)$
Energy
t
$T^+((1-t)\rho_1 + t\rho_2)$
$(1-t)T^+(\rho_1) + tT^+(\rho_2)$
Energy
t
$T^+((1-t)\rho_1 + t\rho_2)$
$(1-t)T^+(\rho_1) + tT^+(\rho_2)$
Energy
t
$T^+((1-t)\rho_1 + t\rho_2)$
$(1-t)T^+(\rho_1) + tT^+(\rho_2)$
Energy
t


$T^+((1-t)\rho_1 + t\rho_2)$
$(1-t)T^+(\rho_1) + tT^+(\rho_2)$
Energy
t

$\int \rho_1 = 4$ and $\int \rho_2 = 6$



$\int \rho_1 = 4$ and $\int \rho_2 = 8$



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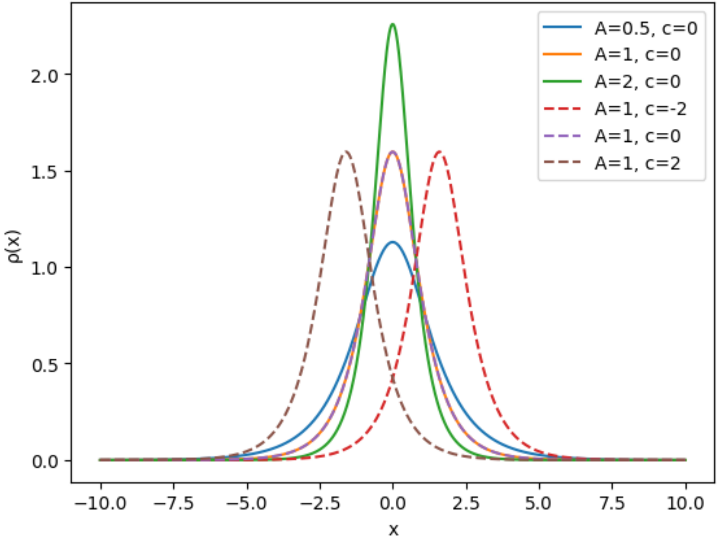
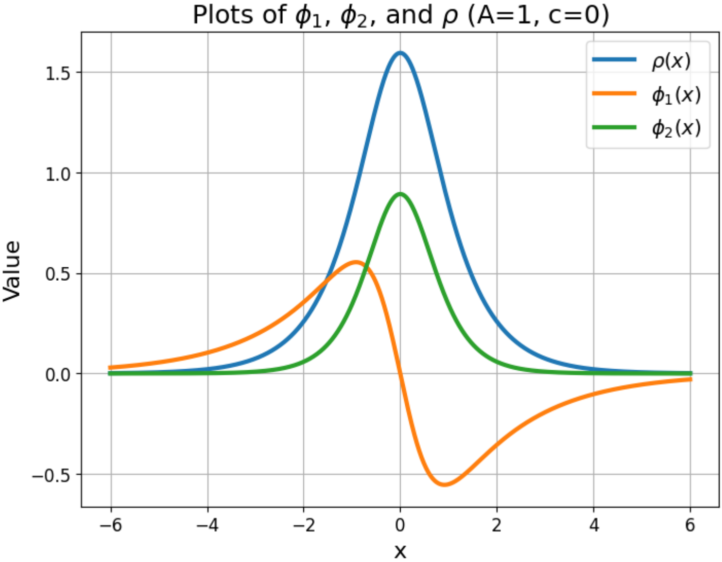
Special class of densities

$$\rho(x) = \frac{4}{\pi}\sqrt{\frac{A\pi}{2}}\,\mathrm{sech}\left(\sqrt{\frac{A\pi}{2}}x + c\right), \quad A > 0, c \in \mathbb{R}.$$

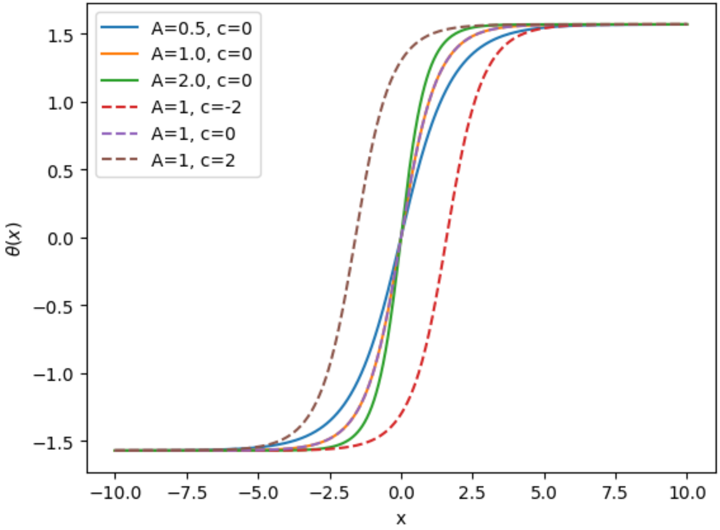
- **Boundedness:** $0 < \rho(x) \leq \frac{4}{\pi}\sqrt{\frac{A\pi}{2}} \quad \forall x \in \mathbb{R}.$
- **Decay at infinity:** $\rho(x) \to 0 \quad \text{as } x \to \pm\infty.$
- **Area under curve:** $\int_{-\infty}^{\infty} \rho(x)\, dx = 4.$
- **Translation invariance:** The parameter c shifts the profile without changing its shape.


A=0.5, c=0
A=1, c=0
A=2, c=0
A=1, c=-2
A=1, c=0
A=1, c=2
ρ(x)
2.0
1.5
1.0
0.5
0.0
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x

Solution


Plots of ϕ_1, ϕ_2, and ρ (A=1, c=0)
$\rho(x)$
$\phi_1(x)$
$\phi_2(x)$
Value
x
1.5
1.0
0.5
0.0
−0.5
−6
−4
−2
0
2
4
6

Angle

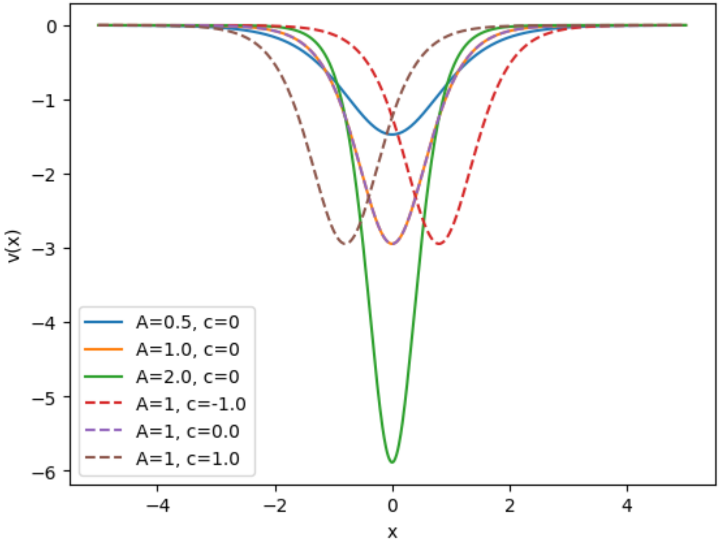

A=0.5, c=0
A=1.0, c=0
A=2.0, c=0
A=1, c=-2
A=1, c=0
A=1, c=2
θ(x)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x

Energy and potential

$$T^{+}(u) = \frac{1}{2}\int_{-\infty}^{\infty} \left(\frac{du}{dx}\right)^{2} dx + \frac{\pi^{2}}{32}\int_{-\infty}^{\infty} u^{6}\, dx$$

$$v(x) = -\frac{15}{16}A\pi\,\mathrm{sech}^{2}\left(\sqrt{\frac{A\pi}{2}}\,x + c\right)$$

Modified Pöschl-Teller potential



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Key Contributions and limitation

- Analytically extended the extends the non-interacting kinetic energy functional
- Overcame the instability from the adjoint problem due to $\rho = \sum_i |\phi_i|^2$ using a parametric representation
- Developed a new approach for explicitly computing the non-interacting kinetic energy functional
- Validated the theoretical using a 1D Kohn-Sham system with randomly generated 1D potentials

Limitations

- Numerical validations are limited to 1D cases.
- No guaranteed scheme for achieving global minimization.
- The minimisation problem may not scale efficiently with increasing number of electrons.

Future Directions

- Extend numerical validation to 3D systems.
- Develop an effective global optimization approach, or provide a rigorous argument that the chosen initialization leads to convergence to a global (or near-global) minimizer.
- Leverage this framework to generate well-controlled and more uniformly distributed training data for machine learning.